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Interfacing an LCD1602 with Arduino Nano, Uno, Mega, and Leonardo.

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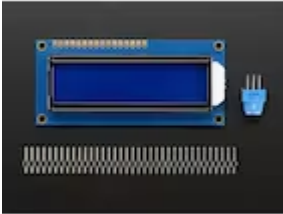
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Arduino IDE

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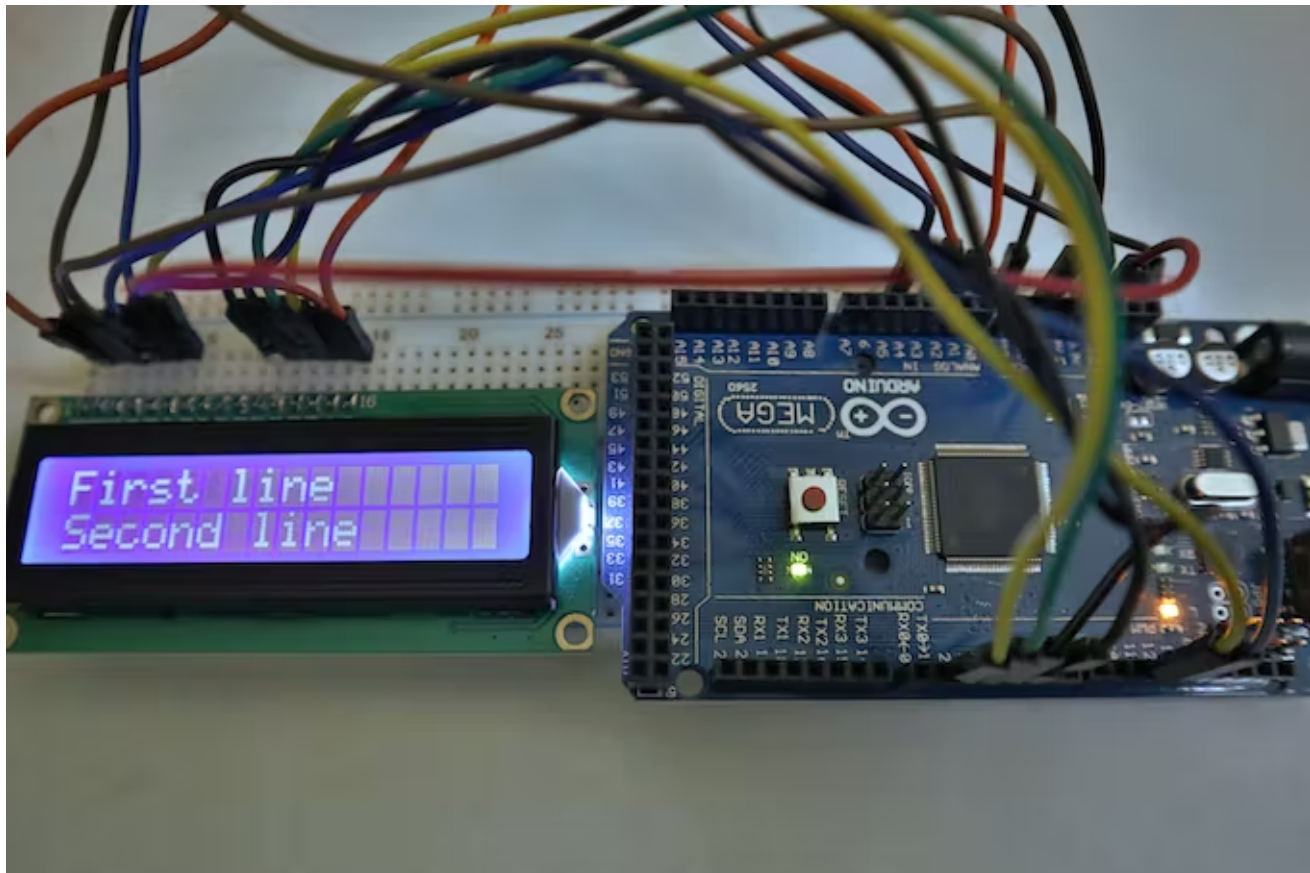
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ABOUT THIS PROJECT



Introduction

In this getting started with LCD module tutorial, we are going to learn interfacing an LCD module with popular development boards like Arduino Nano, Uno, Mega, and Leonardo.

[< PREVIOUS](#)[NEXT >](#)

Hardware Used

- 16x2 LCD module
- Arduino UNO
- Arduino Nano
- Arduino Leonardo
- Arduino Mega

Software Used

- Arduino IDE

Libraries Used

- LiquidCrystal Library - Will available in Arduino IDE by default - For the first method.

A Little Bit About LCD Modules...

LCD modules are coming in different colors and sizes with a different number of displayable characters. the most commonly used one is LCD1602 which can display 16 characters in each line, that is a total of 32 characters. Some other sizes are,

- LCD1604 - 16 char / 4 line
- LCD2004 - 20 char / 4 line
- LCD1602 - 16 char / 2 line
- LCD1601 - 16 char / 1 line etc...

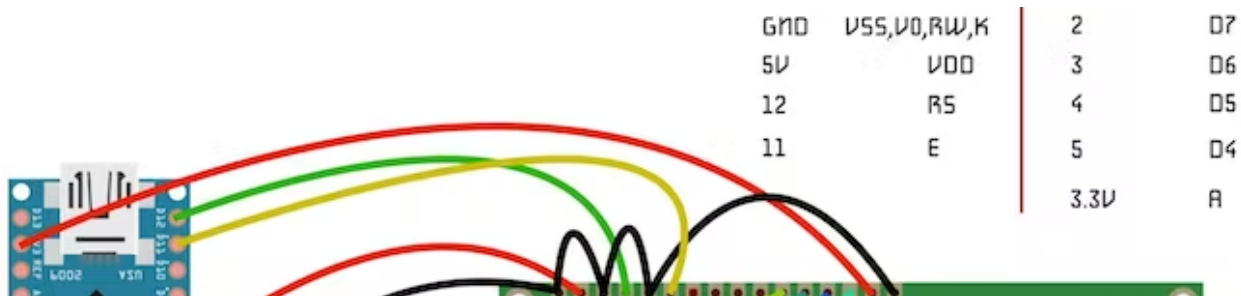


PIN Description

Pin Number	Name	Description
1	VSS (0v)	Ground Potential
2	VDD (5v)	Positive Voltage
3	VO (Contrast)	Contrast adjustment; 0v: Max contrast; 5v: Min contrast
4	RS	Register Select; 0: Instruction Register 1: Data Register
5	RW	Read Write Select pin 0: Write mode; 1: Read mode;
6	E	A Enable Pin To enable the LCD Module
7	D0-D7	LCD Data Bus line. They are responsible for the parallel data transfer.
15	LED+ (A)	Back Light Source LED Anode
16	LCD- (K)	Back Light Source LED Cathode

We are interfacing the LCD1602 with Arduino using the data pins of the LCD module. The same code and circuit will work for all Arduino and compatible boards. Connect the circuit as shown in the below diagram.

Circuit



#	Arduino	LCD1602
1	GND	VSS,V0,RW,K
2	5V	VDD
3	12	RS
4	11	E
5	5	D4
6	4	D5
7	3	D6
8	2	D7
9	3.3V	A

- First connect the ground of Arduino to the VSS of the LCD.
- Then connect the V0 of the LCD to the ground for full contrast
- Then connect RW to the ground for selecting write mode
- Then connect K, which is the ground of backlight LED also to the ground.
- Then connect the 5V of Arduino to the VDD of the LCD module.
- Then connect the digital pin 12 of Arduino to the RS of LCD module.
- Then connect the digital pin 11 of Arduino to the E of LCD module.
- Then connect the digital pin 5 of Arduino to the D4 of LCD module.
- Then connect the digital pin 4 of Arduino to the D5 of LCD module.
- Then connect the digital pin 3 of Arduino to the D6 of LCD module.

- Then connect the digital pin 2 of Arduino to the D7 of LCD module.
- And finally connect the 3.3V of Arduino to the A of LCD which is the anode of backlight LED.

Video

Interfacing LCD1602 With Arduino Without I2C



If you like my work please support me by subscribing (<http://bit.ly/2R95CrQ>) to my channel here (<http://bit.ly/2R95CrQ>)

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CODE



Gist

1 [//www.diyusthad.com](http://www.diyusthad.com)

```
2  #include <LiquidCrystal.h>
3
4  const int rs = 12, en = 11, d4 = 5, d5 = 4, d6 = 3, d7 = 2;
5  LiquidCrystal lcd(rs, en, d4, d5, d6, d7);
6
7  void setup() {
8    lcd.begin(16, 2);
9    lcd.print("First line");
10   lcd.setCursor(0,1);
11   lcd.print("Second line");
12 }
13
14 void loop() {
15 }
```

view raw

(<https://gist.github.com/najadt/de239adc7ace3d840a20f50eca39f0f9/raw/220c4ea0b1ab07e2d2a6e3c11f9f0a425f71a685/LCD1602.ino>)

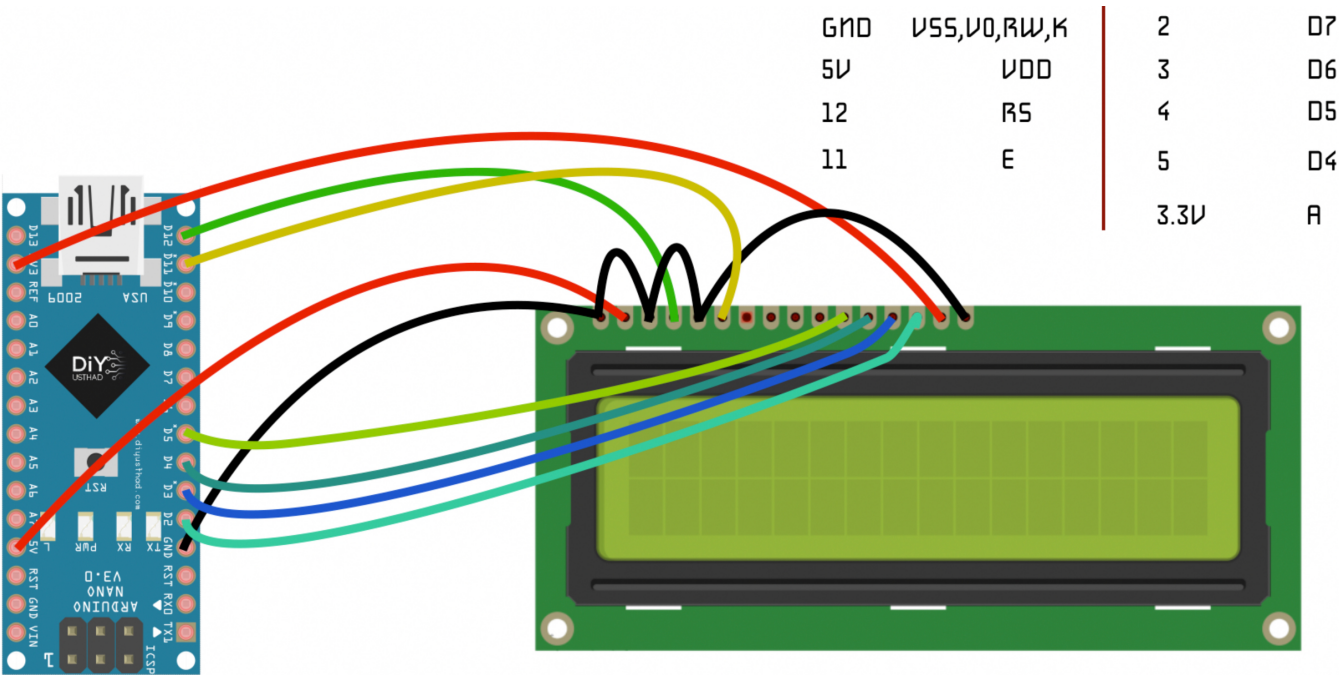
LCD1602.ino (<https://gist.github.com/najadt/de239adc7ace3d840a20f50eca39f0f9#file-lcd1602-ino>) hosted with ❤ by GitHub (<https://github.com>)

SCHEMATICS



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 **jbrahy (/projecthub/jbrahy)**
a year ago

(/projecthub/jbrahy) Thanks for the tutorial. I've tried rewiring it a couple times but I keep getting completely white blocks instead of showing any text. I'm assuming this is related to the contrast. How do I adjust that?

**jbrahy (/projecthub/jbrahy)**

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(/pr
ject Turns out I had a bad wire.

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**obrentz (/projecthub/obrentz)**

9 months ago

(/pr
ject Which wire was it?

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**snyderdan46 (/projecthub/snyderdan46)**

7 months ago

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ject I had to add a trimmer pot to the V0 pin, which controls the contrast. There is a
Hub/ VERY small range between white blocks and no display at all.

snyd

erda

n46)

**wgyhffd357 (/projecthub/wgyhffd357)**

2 months ago

(/pr
ject added a 2k resistor to v0 pin

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**n1kk (/projecthub/n1kk)**

a month ago

(/pr
ject I tried adding 2k resistor to V0 but I still have a line of white squares. Also tried
Hub/ adding a 10 potentiometer and tune contrast with that, still no text.

n1kk

)

**n1kk (/projecthub/n1kk)**

a month ago

(/pr

Made it work, for some reason if I wire RS and E to D12 and D11 it just doesn't work.
hardwired it to be rs = 4, en = 5, d4 = 6, d5 = 7, d6 = 8, d7 = 9 and now I see text
n1kk
)

AUTHOR



(/projecthub/najad)

Najad (/projecthub/najad)

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Interfacing the Onion Omega2 and Arduino Uno via UART (/projecthub/supersonnic/interfacing-the-onion-omega2-and-

by Shervin Oloumi (/projecthub/supersonnic)

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